

The Pile Game

Week 1 · Grades 4–5

Name: _____

Mathematicians call this game **Bulgarian solitaire**. It became famous through a magazine column in 1983, and it hides a genuine theorem—one whose proof was only found in 1982. You will find the theorem yourself today, and prove the parts of it that can be proved in an hour.

The rule

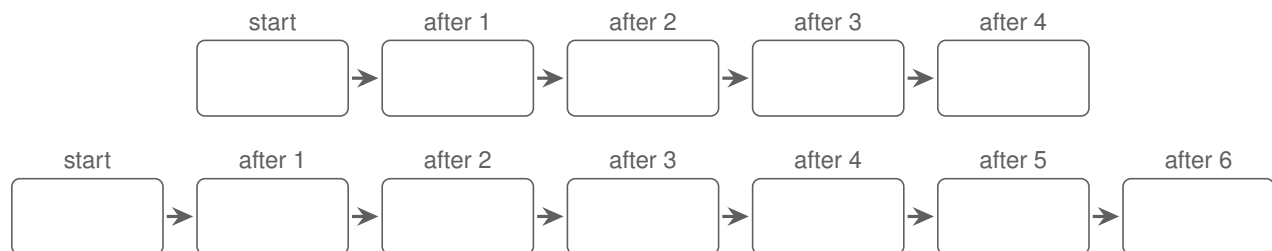
Take one counter from **every** pile. Put those counters together: they make a **new pile**. A pile that had only one counter is **gone**. That is one **move**. Repeat forever.



The dark counters are the ones being taken; after the move they form the dark pile. **Notation:** a *position* is the list of pile sizes, biggest first, like **3 1 1**. Where the piles sit does not matter, only their sizes. If the piles are the same after a move, the position is **stuck** (a *fixed point*).

Warm-up: six counters

Play from **6** and from **2 2 1 1**. Write every position.



What do you notice? Compare with your table.

The questions we are chasing

1. Which numbers of counters end up **stuck**, whatever the start? What is special about those numbers?
2. **Why** is the stuck position stuck—and is there ever a different stuck position?
3. What happens with the other numbers of counters?
4. Which positions can **never** be reached from anywhere?
5. How **long** can the road to the stuck position be?

1. Survey: one to ten counters

Split the numbers among your table. For each number of counters, play from at least two different starts and record what happens: stuck at which position, or looping through which positions?

counters	starts you tried	stuck at ...	or loops through ...
1			
2			
3			
4			
5			
6			
7			
8			
9			
10	whole group		

1. Which numbers of counters get stuck? _____ What do these numbers have in common? Predict the next two.

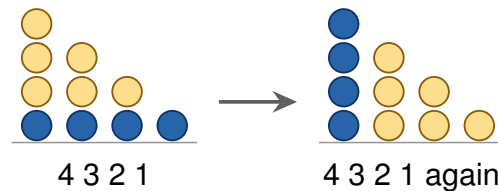
2. Lay each stuck position out as columns, tallest first. What shape do you see? Why does that explain which numbers get stuck?

3. For 8 counters, did everyone at your table find the same loop? _____

4. **Extra.** How many different positions are there altogether with 4 counters? _____ With 5? _____ With 6? _____ With 7? _____

2. Why?

(a) **Why is the staircase stuck?** Lay **4 3 2 1** out as columns. Taking one counter from every pile is the same as taking the whole **bottom row**.



Explain in your own words why the move gives back exactly the same shape. (How many counters are in the bottom row? Why?)

(b) **Is anything else stuck?** Suppose some position is stuck: after one move the piles are exactly the same as before. Say it has m piles. Fill in the blanks.

The new pile has _____ counters, because there were m piles. Every old pile got smaller, so the biggest pile after the move must be the _____ pile. The position did not change, so its biggest pile has _____ counters.

The second-biggest pile after the move is the old biggest pile with one counter removed, so it has _____ counters. The position did not change, so its second-biggest pile has _____ counters.

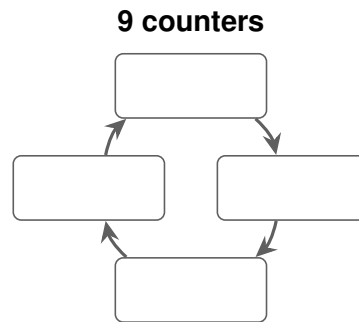
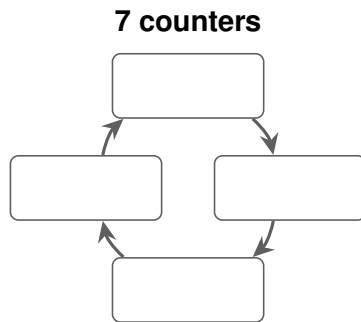
Continue: each pile is _____ smaller than the one before. There are m piles, so the sizes are m , $m - 1$, ..., _____. The position is a _____!

(c) Use (b) to explain to your table why a game with 8 counters can **never** get stuck.

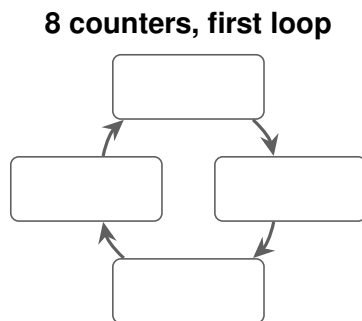
(d) **Why must every game eventually repeat?** With 10 counters there are exactly 42 different positions. Suppose you play 42 moves from any start. Why must some position appear twice? And once a position appears a second time, what happens next—and why?

3. Loops

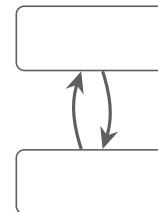
7 counters: play from **7** and from **2 2 2 1**; write the loop you reach (4 positions). **9 counters:** one loop again; write it.



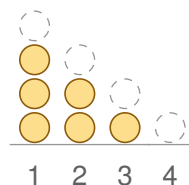
8 counters: there are **two** different loops. Write both. (Starts to try: **8**, **5 3**, **2 2 2 2**.)



8 counters, second loop



1. Make each loop position with counters, laid out as columns, right next to the staircase **3 2 1**. Every loop position is the staircase **plus some extra counters**. How many extras for 7, for 8, for 9? Where are the extras allowed to sit? (There are exactly four places; the picture shows them dashed.)



2. Follow **one** extra counter through a loop. After each move, which place is it in? Describe the pattern, and explain it with the “bottom row stands up” picture.

3. Why does 8 have a loop of length 2? (Which places do its extras sit in?)

4. **Predict, then test.** 11 counters is the staircase **4 3 2 1** plus one extra. How many positions are in the loop? _____ 12 counters: how many loop positions altogether _____, and how many different loops? _____

4. Backwards

1. Find a position that leads to **3 3** in one move. _____
2. Find **all** positions that lead to **3 2**. (There are two.) _____
3. Find all positions that lead to **4 3 2**. (There are three!) _____
4. Now find a position that leads to **2 2 1 1**. Keep trying for a while...

Nothing leads to 2 2 1 1. Here is why. Fill in the blanks.

Suppose a position was just produced by a move. Its new pile has as many counters as there were piles _____ the move. Every other pile lost a counter, so it came from a pile with at least _____ counters—and the number of those old piles is at most the size of the new pile.

So after a move: (number of piles) $\leq$ (size of the new pile) $+ 1 \leq$ (size of the biggest pile) $+ 1$.

2 2 1 1 has _____ piles, but its biggest pile has only _____ counters. So no move produces it.

A position that no move can produce is called a **Garden of Eden**: you can start there, but never return.

5. State the rule for spotting a Garden of Eden: a position is a Garden of Eden exactly when _____ (in words).

6. List all Gardens of Eden with 6 counters (there are 3) and with 7 counters (there are 5).

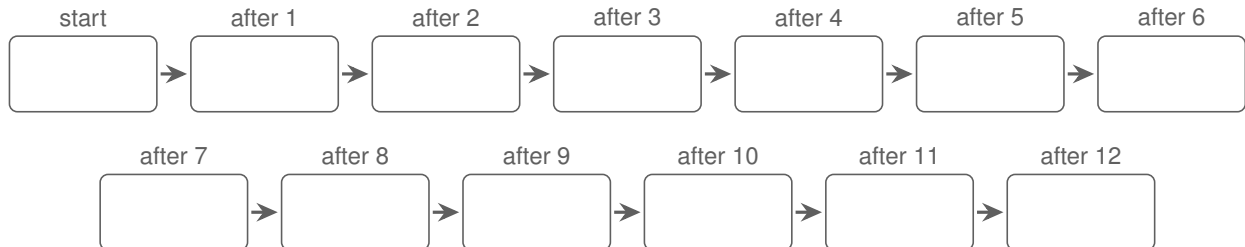
7. **The backwards recipe.** Take a position with p piles. Choose a pile of size s with $s \geq p - 1$. Remove it, add one counter to every other pile, then add $s - (p - 1)$ piles of one. Try it on **4 2 2**. Check by playing forwards. Why does the recipe work? Why must every parent come from this recipe?

8. Can a position have four different parents? Find one, or explain why not. (Hint: how many different pile sizes are at least $p - 1$?)

5. The longest road

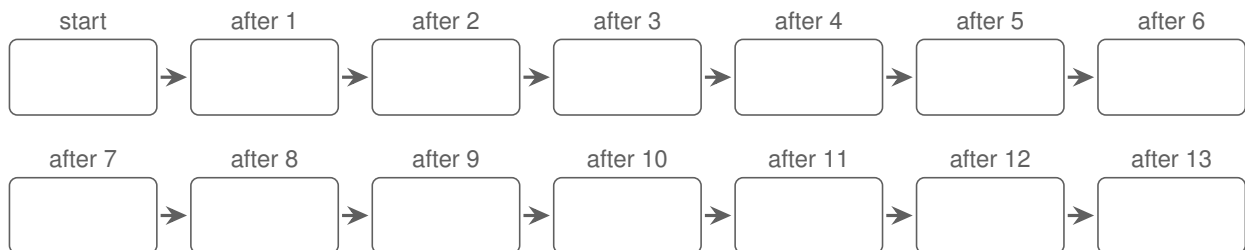
With 3 counters the longest road to the staircase has 2 moves: **1 1 1** → **3** → **2 1**. With 6 counters the longest road has 6 moves (you found it: from **2 2 1 1**). With 10 counters the record is **12 moves**.

- Find a start with 10 counters whose road is 12 moves long. Hints: the start must be a Garden of Eden (why?), and it has five or six piles.



- Which starts with 10 counters reach **4 3 2 1** in exactly one move? _____ In exactly two moves? _____
- The records so far: 2 moves for the 2-step staircase, 6 for the 3-step, 12 for the 4-step. What is the pattern? Predict the record for 15 counters (the 5-step staircase). _____

- Mathematicians proved that with the k -step staircase the road is never longer than $k \times (k - 1)$ moves—and for 15 counters some start really does take 20. Suppose you start 15 counters as one pile, **15**. Is the road long or short? Guess, then play it.



- Take home.** Find a start with 15 counters whose road is 20 moves long. Then explain to someone at home why every game with 15 counters ends at **5 4 3 2 1**—or admit which part you cannot prove yet. (Nobody could until 1982.)